

Improved Time Series Classification with Representation Diversity and SVM

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Abstract—Time series classification is an important task in data mining that has been traditionally addressed with the use of similarity-based classifiers. The 1-NN DTW is typically considered the most accurate model for temporal data. Nevertheless, some authors have recently proposed ingenious alternatives to the 1-NN DTW by using diversity of time series representation or by using DTW for feature extraction. In this paper, we explore diversity of time series representations and distance functions to obtain distance features, which in turn are used to train an SVM model. We argue that the scientific community has largely neglected a vast body of unconventional distance functions, and we present empirical evidence that distance features are better than the 1-NN DTW with respect to classification accuracy.

I. INTRODUCTION

Time series classification is a ubiquitous task across a wide range of applications in academia, industry, and government. Recent empirical evidence [1]–[3] has shown that the 1-nearest neighbor classifier (1-NN) presents very competitive results when allied to Dynamic Time Warping (DTW), a distance measure with local time-scaling invariance.

The 1-NN classifier is based on a very simple approach. From practical and theoretical standpoints, there are no definitive arguments to support that such a particular algorithm is better suited for time series data than any other classification model. In our opinion, the supremacy of 1-NN can be better understood by the inability of traditional machine learning approaches to capture the dependencies among the non-i.i.d.¹ observations. The main challenge of learning classification models with time series data is to find meaningful features. Although a single time series observation may be cast as a feature value, its representation power is diminished since it is taken independently from its neighboring observations. In order to be relevant, time series features should be obtained from sequences or subsequences of observations.

A straightforward alternative to extracting features from temporal data is using similarity-based classifiers. In the case of time series, the use of distance functions that promote invariances such as warping, amplitude, offset, and complexity allows robust comparison of time-oriented sequences by ignoring some undesired properties of the data [2]. We believe that the competitive empirical results obtained by 1-NN can be better explained by the use of these invariances than by the inductive bias of the algorithm per se.

¹Independent and identically distributed. Time-oriented data is not independent given the typical auto-association among observations.

Recently, Kate [4] has empirically shown that a state-of-the-art classifier, Support Vector Machine (SVM), allied to DTW can provide excellent experimental results, outperforming 1-NN with DTW. In that work, the DTW distance is used to provide features for a standard machine learning classifier. The procedure is to describe a given example $\mathbf{x}$ as a sequence of DTW distances from $\mathbf{x}$ to a set of reference time series, usually an entire training set.

In this paper, we present an extension of [4] by adding representational diversity to the feature set. The main intuition behind our contribution is that not all features can be properly represented in the time domain. An example is the well-known frequency representation that allows describing a time series by a sequence of increasing frequency components, known as frequency spectrum.

We avail of a body of time series representations proposed in the literature, such as frequency, wavelets, and autocorrelation to build a diverse set of distance features. We show that these features allied to an SVM classifier can outperform the 1-NN-DTW and also the SVM classifier with DTW features as proposed by [4].

The remainder of this paper is organized as follows: we introduce time series and distance functions in Section II. In Section III, we introduce the SVM and 1-NN classifiers. We cover DTW features and present our main contribution in Section IV. Our experimental methodology is explained in Section V, and our results are discussed in Section VI. Finally, we present conclusions and future work ideas in Section VII.

II. TIME SERIES AND DISTANCE FUNCTIONS

Let a time series S of length m be an ordered set of values $S = (s_1, s_2, \dots, s_m)$. Each $s_t \in \mathbb{R}$ is an observation of S at the instant t .

A distance function $f(S, Z)$ is an estimate of the similarity or dissimilarity between time series. Distance functions are at the heart of most data mining tasks, and the Euclidean distance is arguably the most known and used distance function for temporal data. It is a particular case of a Minkowski distance (Equation 1) when $p = 2$.

$$d_{Mink-p}(S, Z) = \left(\sum_{t=1}^m |s_t - z_t|^p \right)^{\frac{1}{p}} \quad (1)$$

Distances such as the Euclidean are *lock-step measures*. Lock-step measures give the distance between series from an alignment between pairs of observations that must be identically placed in the time axis, as exemplified in Figure 1 (left). This results in the fact that lock-step measures often lead to the conclusion that series with small distortions in the time axis, a phenomenon known as *local warping* [5], are more distant from one another than they actually are.

Conversely, an elastic measure gives the distance between series by means of a dynamic alignment of their observations, as shown in Figure 1 (right). For temporal data mining, the Dynamic Time Warping (DTW) is certainly the most explored elastic distance. It promotes an alignment that minimizes the observed distance by specifically reducing the effects of local warping. For time series classification, the 1-Nearest Neighbor classifier has been repeatedly reported to be extremely efficient when associated with the DTW distance [2]. The association of the 1-NN classifier with DTW is so common that it is usually referred to by a name of its own – 1-NN DTW.

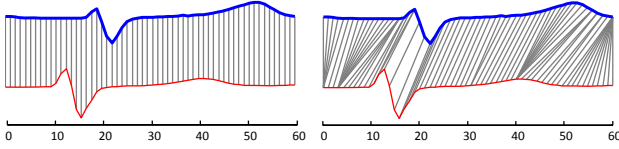


Fig. 1. Alignment between observations promoted by a lock-step distance (left) and by the DTW distance (right) for the same pair of time series.

The warping promoted by the DTW distance is subject to several constraints [3]. However, it does not impose restrictions on the distance between matching observations on the time axis. This sometimes leads to pathological warping where one series is excessively warped to minimize the distance against the other, causing the distance to be underestimated. One way to avoid this issue is to use warping restrictions, such as the Sakoe-Chiba window [6]. It consists in applying a sliding window while matching the observations. Two observations can match only if they fall simultaneously in the sliding window. From a practical point of view, Sakoe-Chiba windows restrict the amount of stretching that may be applied on the time axis.

III. CLASSIFICATION OF TIME SERIES

The task of classification consists in assigning a class label C_x to an instance x of unknown label. For example, given a time series x that represents a single heartbeat within an ECG, its class label C_x may indicate whether this beat is regular or anomalous.

The major approaches for time series classification are feature-based and similarity-based classification. The first case consists in extracting relevant features from the series in order to build a classification model, such as the Support Vector Machine (SVM). The SVM is an instance-based classifier that projects the training samples into a space of higher dimensionality, where it assumes to have a representative

layout of the original space. In this projection, the SVM attempts to find the hyperplane that best separates the classes, effectively dividing the decision space into two subspaces. When classifying a new sample, the SVM projects its features into the same high-dimensional space and verifies on which subspace the projected instance “falls”, and assigns it the class label associated with that subspace.

Similarity-based classification, on the other hand, depends directly on a similarity or distance function. A simple similarity-based algorithm is the k -nearest neighbors (k -NN) classifier. To classify a new instance x , the k -NN classifier first finds the k training examples that are most similar to x – i.e., the nearest neighbors of x . Then, it uses the labels of the nearest neighbors to estimate the label of x . The 1-NN DTW is arguably the most used approach for time series classification [1].

Recently, some authors have proposed ingenious methods that use ensembles [7] or complex classification models to gain on the overall accuracy of the 1-NN DTW. In this paper, we have built on the idea presented by Kate [4], which consists in using the DTW distance to extract features known as DTW features in order to train an SVM classifier. We augment this general idea by employing different representations of time series and distance functions.

IV. DISTANCE FEATURES

The main challenge of time series classification is to find meaningful features. While “traditional” machine learning techniques usually ignore or actively seek to remove correlated variables, the correlation among observations in temporal data is essential to the discrimination of the classes. Therefore, efficient features for temporal data must be extracted from whole sequences or subsequences of time series. Examples of such features include shapelets [8], which are distinctive subsequences of each class, and mel-frequency cepstral coefficients [9], which are extracted from the frequency representation of the whole sequence. However, though certain types of features are known to work well in certain domains, it is usually acknowledged that no feature provides discriminative power that generalizes to different application domains as well as the DTW distance [1].

The use of distance functions that promote invariances such as warping, amplitude, offset, and complexity allows robust comparison of time-oriented sequences by ignoring some undesired properties of the data [2]. This approach has been traditionally associated with similarity-based classification models, such as the k -NN. In the past decades, most of the effort in time series classification has been devoted to the 1-NN DTW classifier.

Recently, Kate [4] has proposed DTW features as a method to use the DTW distance to extract generic features for time series. DTW features are calculated as the DTW distances from one series to a set of reference series. Essentially, they encode how each example fits into its neighborhood. With a reasonable set of reference data, DTW features may be used

to incorporate the invariances promoted by the DTW distance into any classification model.

The procedure to extract DTW features is fairly simple. Let S be a time series of interest and $D = \{P_i\}_{i=1}^n$ a set of reference series. For instance, D may be the entire set of training instances. Then, the DTW features of S are the feature vector $S^{DTW} = (DTW(S, P_1), DTW(S, P_2), \dots, DTW(S, P_n))$.

Besides DTW features, [4] also used the Euclidean distance to obtain ED features, and associated DTW with Sakoe-Chiba windows to extract DTW-R features. In this work, we build on those ideas by employing diversity of time series representations, associated with a richer set of distance functions to extract features, which we refer to as *distance features*.

A set of distance features for S is extracted with respect to a time series representation R and a distance function f . Let $\varphi_R(S)$ describe the transform of S according to R . Then, the distance features of S with respect to R and f are the feature vector $S^{R,f} = (g(1), g(2), \dots, g(n))$, where we use the notation $g(i) := f(\varphi_R(S), \varphi_R(P_i))$ for simplicity. For instance, features extracted from the amplitude spectrum representation with Euclidean distance would be given by $\varphi_{AMP}(S) = AMP(S)$, where $AMP(S)$ is the amplitude spectrum representation of S , and the distance function $f(S, Z) = d_{Mink-2}(S, Z)$.

The notion of using a transform of the temporal data in order to extract features or to promote classification by similarity has been the subject of previous work. Translating the classification problem into an alternative decision space may expose discriminative features that are not clear in the time representation [7], [10]. Arguably the most known and understood example is the Discrete Fourier Transform (DFT) of a time series, from which one may extract frequency features that may be used for tasks such as classification [11] and indexing [12] of temporal data.

There is a huge diversity of time series representations in the scientific literature [13]. In this work, we attempted to choose transformations that actually provide a distinct decision space that is not a summary of the original temporal information. Therefore, we have excluded some classical dimensionality reduction techniques, such as Principal Component Analysis [14], SAX [15] or SAX-based transformations [16]. The transformations we used in our experiments will be presented in the next subsections.

A. Amplitude Spectrum

The Discrete Fourier Transformation (DFT) of a time series gives its decomposition as an ordered set of periodical components of increasing frequency. Each component is described by a complex number that encompasses both its amplitude and phase. As such, the target domain of the DFT is known as the frequency domain [12].

The Amplitude Spectrum is a representation that derives from the DFT. It is obtained from the absolute value of each Fourier coefficient and is closely related to the Power Spectrum. The Amplitude Spectrum gives the amplitude of each frequency component and may be used to expose periodical

trends in the time domain, even if the original series contains noise. For problems in which the data tends to be very long and noisy, or when it expresses phenomena that are best defined by frequency components (e.g., voice or speech recognition), the Amplitude Spectrum representation can provide an efficient decision space for classification.

B. Discrete Wavelet Transformation

An important observation about signal processing and, oftentimes, temporal data mining in general, is that low-frequency components are usually lasting within signals while high-frequency components tend to exist for shorter periods. It follows that low-frequency components require less certainty in time information while high-frequency components require higher certainty in time information [17].

Wavelet transforms arrive at this scenario as a form of Multi-Resolution Analysis. The wavelet transformation is performed with a sliding window, defined by a function called *mother wavelet*. The wavelet is scaled up for lower frequencies, so that these components are extracted from wider subintervals of the signal and, therefore, favor frequency information over time information. Conversely, the wavelet is scaled down for higher frequencies, favoring time over frequency. The resulting transform is a time-frequency representation of the original signal or time series.

The simplest DWT variation is the Haar transform [18]. This technique is widely applied in many different applications domains. For example, it may be used with machine learning methods for spoken digits recognition [19] and matching of stock time series [20].

C. Autocorrelation Function

The autocorrelation of a time series measures its predictability at any instant from its previous observations. A highly autocorrelated time series is indicative of a very deterministic process, while true noise shows no autocorrelation for pairs of distinct observations.

The autocorrelation transform of a time series is obtained from its sampled autocorrelation coefficients [21]. Autocorrelation coefficients indicate how much the observations of a series are related to each other at different lag values. The distance between two time series in a decision space defined by autocorrelation features is useful to distinguish phenomena produced by different random processes.

V. EXPERIMENTAL EVALUATION

In this work, we have evaluated distance features as attributes for an SVM classifier. As it has become standard practice, we used the benchmark data sets from the UCR Archive [22], which contained 85 data sets from diverse areas at the time this paper was written.

Our major contribution is the use of diversity of time series representation and distance functions in order to extract distance features. The following representations were used:

- Temporal representation (“raw” data);
- AMP: amplitude spectrum representation;

- DWT: Haar wavelets representation;
- ACF: autocorrelative representation.

Each of these representations was used to extract features with different distance functions. Specifically, the following six distances were used, resulting in a total of 24 distinct distance features:

- EUC: Euclidean distance, given by the Minkowski distance when $p = 2$ (Equation 1);
- CHEB: Chebyshev distance, given by the Minkowski distance when $p \rightarrow \infty$ (Equation 2);
- AVG: averaged Manhattan and Chebyshev (Equation 3);
- JAC: Jaccard distance (Equation 4);
- DICE: Dice distance (Equation 5);
- LOR: Lorentzian distance (Equation 6).

We refer to the features used in our evaluation by the abbreviation of the combined representation and distance function. For instance, from the association of amplitude spectrum with Euclidean distance, we obtained the AMP-EUC distance feature. For brevity, we shall omit the representation name for the temporal domain (*e.g.*, EUC instead of TIME-EUC).

$$d_{Cheb}(S, Z) = \max_{t=1}^m |s_t - z_t| \quad (2)$$

$$d_{avg}(S, Z) = (d_{Mink-1}(S, Z) + d_{Cheb}(S, Z))/2 \quad (3)$$

$$d_{Jac}(S, Z) = \frac{\sum_{t=1}^m (s_t - z_t)^2}{\sum_{t=1}^m (s_t^2 + z_t^2 - s_t z_t)} \quad (4)$$

$$d_{Dice}(S, Z) = \frac{\sum_{t=1}^m (s_t - z_t)^2}{\sum_{t=1}^m s_t^2 + \sum_{t=1}^m z_t^2} \quad (5)$$

$$d_{Lor}(S, Z) = \sum_{t=1}^m \ln(1 + |s_t - z_t|) \quad (6)$$

In addition to the aforementioned combinations, the DTW and the DTWR distances were used exclusively in the temporal domain to avoid “warping” frequency or correlation features. We refer to features extracted with the unrestricted DTW distance as DTW and to DTW associated with a Sakoe-Chiba window as DTWR. To find DTWR distance features, we used a DTW implementation based on the UCR suite [23] and set the window size according to the recommendations of [22].

Distance features may be combined in several ways. We explored augmenting the decision space by “concatenating” feature vectors in order to obtain a larger vector. One example of this process comes from the concatenation of DTW and DTWR features, which [4] denoted Feature-DTW-DTW-R and corresponds, in this paper terminology, to the concatenation of DTW and DTWR. We shall use the operator $+$ with some degree of informality to convey the idea of concatenating distance features. In this case, the resulting decision space shall be expressed as DTW+DTWR.

After extracting distance features, we performed 5×2 -fold cross-validation (CV) on the training data to find the best degree for the SVM polynomial kernel. The polynomial kernel for the feature vectors S and Z is $K(S, Z) = (\gamma S^T \cdot Z + C)^d$, where γ and C are sensitivity parameters and d is the polynomial degree. We used Weka [24] and LibSVM [25]

to classify the data and kept $\gamma = 1$ and $C = 1$, which are the default values of LibSVM and also the methodology adopted by [4]. We restricted our scope to linear, quadratic, and cubic polynomials, as polynomials of higher order are much more likely to overfit the training data. For experiments with representation diversity, we also used cross-validation to select the best distance features according to the training data. Our implementation is reproducible and may be found in [26].

VI. RESULTS

Our first experimental setting aimed at reproducing the results of [4]. For each data set, we obtained their DTW+DTWR features and, after tuning the polynomial kernel on the training data, compared against the 1-NN DTW. We conducted a Wilcoxon signed-rank statistical test and verified, at the 95%-confidence level, that DTW and DTWR features perform better than the 1-NN DTW. A summary of wins, ties, and losses for DTW+DTWR is presented in Table I. For the totality of our results, we refer the reader to [26].

Next, we verified if our strategy would be able to select distance features that are competitive against the 1-NN DTW. For each data set, we extracted 24 distance features as explained in Section V. We then proceeded to train several SVM classifiers and estimate their accuracies on the training data. A single combination of distance feature and polynomial degree was selected. We refer to this approach as DISTF-tuned.

When comparing DISTF-tuned with 1-NN DTW, we verified that our approach wins on 54 data sets and loses on 30, as shown in Table I. In case these numbers might suggest that distance features are a weak contender, its strength becomes more clear when we compare the accuracy obtained by both methods in each data set. This is shown by the scatter plot of accuracies in Figure 2. While there are several data sets where 1-NN DTW performs better, distance features win by a much larger margin in most cases. We performed a Wilcoxon signed-rank test and verified that DISTF-tuned wins against 1-NN DTW at the 95%-confidence level. However, when comparing against DTW+DTWR, we were unable to reject the null hypothesis and can not claim that either method is better.

At this point, we wish to draw the reader’s attention to the distinguishing factor of DISTF-TUNED. Though some authors have claimed to surpass the 1-NN DTW with reasonable experimental evidence, those works typically rely on the DTW distance somehow. For instance, the ensemble approach of [7] uses the 1-NN DTW as one of its base classifiers, while DTW features capture the same notion of neighborhood that is exploited by the 1-NN DTW. The addition of representation diversity to the proposal of [4] is remarkable in the sense that

TABLE I
SUMMARY OF WINS, TIES, AND LOSSES FOR THE CLASSIFICATION
APPROACHES DISCUSSED IN THIS PAPER.

	DISTF-tuned	DTW+DTWR	1-NN DTW
DTW+DISTF-tuned	47/13/25	52/7/26	67/1/17
DISTF-tuned		38/5/42	54/1/30
DTW+DTWR			55/1/29

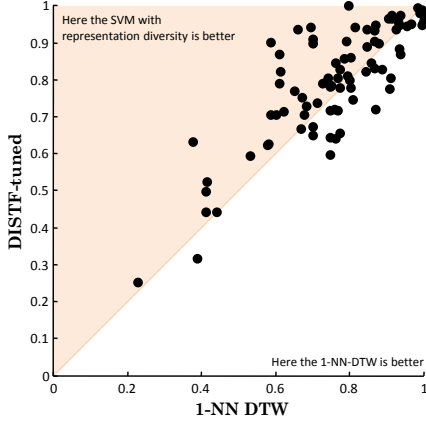


Fig. 2. Accuracy plot for DISTF-tuned versus 1-NN DTW. Points above the diagonal indicate the method on the Y-axis is better in a particular data set.

it overcomes the 1-NN DTW without making use of DTW distance.

Nevertheless, we do not make any attempt to reject neither the 1-NN DTW classifier nor approaches that associate 1-NN DTW with other techniques. In fact, we hypothesized that the best results would be given by the association of DTW features and representation diversity. To verify this, we associated all 24 distance features with DTW, producing several composite decision spaces such as DTW+EUC, DTW+AMP-EUC, DTW+AMP-CHEB, etc. We used those features to construct several SVM classifiers, which we then proceeded to evaluate on the training data with 5×2 -fold CV. The best combination of features and polynomial degree was selected for each data set and experimented on the test data. We shall refer to this strategy as DTW+DISTF-tuned.

As seen in Table I, DTW+DISTF-tuned wins against the 1-NN DTW on 67 data sets, and loses on 17. A Wilcoxon signed-rank hypothesis test was conducted, and it enables us to assert that our approach is better than the 1-NN DTW at the 95%-confidence level. To best demonstrate the magnitude of the difference between the two methods, a scatter plot of accuracies is presented in Figure 3.

The next question is whether associating DTW features with DISTF-tuned is better than the association of DTW and DTWR features. As shown in Table I, DTW+DISTF-tuned wins against DTW+DTWR on 52 data sets, ties on 7, and loses on 26. According to a Wilcoxon signed-rank test, associating DTW features with distance features is better on the 95%-confidence level. A scatter plot of accuracies between these approaches is shown in Figure 4.

Finally, we have performed a Friedman hypothesis test of the four previously discussed approaches. The Friedman test is the non-parametric equivalent of repeated ANOVA tests [27], and it is appropriate for comparing multiple classifiers when the distribution is not known. The Friedman test rejected the null hypothesis that all algorithms are comparable for 85 data sets at the 95%-confidence level. We therefore conducted

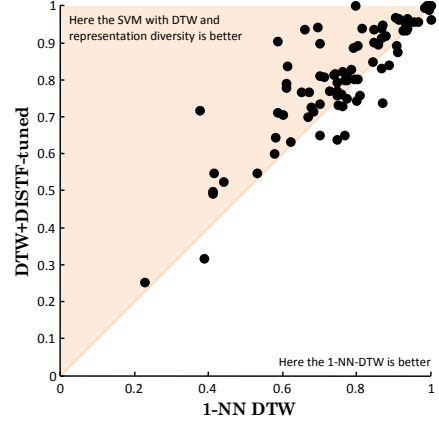


Fig. 3. Accuracy plot for DTW+DISTF-tuned versus 1-NN DTW. Points above the diagonal indicate the method on the Y-axis is better in a particular data set.

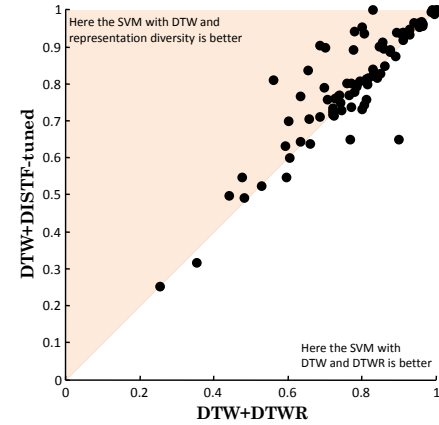


Fig. 4. Accuracy plot for DTW+DISTF-tuned versus DTW+DTWR. Points above the diagonal indicate the method on the Y-axis is better in a particular data set.

a post-hoc Nemenyi test to verify which algorithms were significantly different. A critical difference graph is presented in Figure 5.

The graph in Figure 5 shows the average ranks obtained by all approaches. Smaller ranks are better. For 85 data sets and four algorithms, mean rank differences above 0.51 indicate statistically significant difference. This result confirms all previously discussed Wilcoxon tests, and it allows us to conclude, at the 95%-confidence level, that our approach is better than both the 1-NN DTW classifier and the SVM classifier with DTW+DTWR features.

VII. CONCLUSIONS AND FUTURE WORK

In spite of recent efforts, we believe the power of alternative representations for time series classification deserves more attention from the machine learning community.

In this sense, we have presented a strategy to combine the concepts of representation diversity with DTW features in

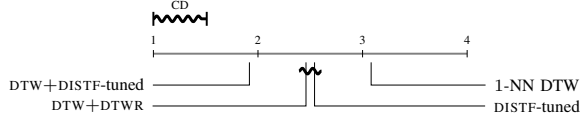


Fig. 5. Critical difference graph for the classification approaches studied in this paper. The critical difference of mean ranks is 0.51. Methods joined together by a thick line indicate rank difference under that value. Methods not connected are considered to behave differently at the 95%-confidence level. Lower ranks are better.

an approach we named distance features. Distance features captures the concepts of neighborhood provided by the different combinations of time series representations and distance functions and express them as generic features that may be used to train any classification model.

In this paper, we have experimented using distance features to train SVM models. We have performed extensive and careful experimental evaluation and showed that this approach is better than not using representation diversity, and that it is also more accurate than using DTW distance only – either in association with the 1-NN classifier or as a feature-extraction method to power more robust classification models such as the Support Vector Machine.

We suggest that further study of distance features should be conducted, including studies with DTW features, SAX features and features extracted on other domains of representation and by distance functions not included in this study. For instance, we intend to include recurrence plot-based representations and compression-based distances. In our opinion, there is a vast body of time series representations and unconventional distance functions that should be explored by the temporal mining community.

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